Dr. Alexander Sehlke

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Summary

Planetary Geologist at NASA Ames Research Center/BAER Institute, with focus on scientific exploration of the solar system through returned sample analyses, fieldwork, space flight hardware development, and NASA-led science exploration missions. Serving as Principal Investigator (PI) and Co-Investigator (Co-I) on multiple research awards funded by NASA's Science Mission Directorate (SMD).

Education

PhD **University of Missouri, Columbia MO**, Geological Sciences — Jan 2011 – Dec 2015
- Minor in College Science Teaching

Dipl **Leibniz Universitaet Hannover, Germany**, Geosciences — Oct 2005 – Jan 2011
- Minor in Material Sciences

Professional Appointments

NASA Ames Research Center/ BAER Institute, Research Scientist — Moffett Field, CA, USA — Jan 2019 – present
- *Apollo Lunar Sample Analysis*: Investigating the thermochronology of the lunar surface and subsurface using thermoluminescence, with a focus on cold trap prospecting in support of NASA's Artemis program.
- *Planetary Analog Research*: Leading field and laboratory studies of terrestrial analog sites to better understand planetary surface processes on the Moon and Mars.
- *Spaceflight Instrumentation*: Designing, developing, and field-testing advanced scientific instruments for robotic and crewed planetary exploration.
- *NASA VIPER Mission*: Instrument Scientist for the Volatiles Investigating Polar Exploration Rover (VIPER), contributing to hardware development and science operations.
- *NASA Ames Vertical Gun Range (AVGR)*: Serving as Science PI for NASA's premier hypervelocity impact facility, enabling experimental studies of planetary surface processes and impact dynamics.

NASA Ames Research Center/ USRA, Postdoctoral Researcher — Moffett Field, CA, USA — Feb 2016 – Jan 2019
- Planetary Analog Research: Volcanic Terrains on Earth as Analogs for the Moon and Mars
- Instrument Development for Human Space Exploration Missions

Awards and Honors

NASA Ames Honor Award
- Contractor Employee in 2020
- Partnership RESOURCE Research Program in 2020
- Team/Group BASALT Research Program in 2020
- Team/Group FINESSE Research Program in 2018

Committees and Assignments

SSERVI NASA Exploration Science Forum, Science Organization Committee — Mar 2025 – May 2025

SSERVI NASA Exploration Science Forum, Session Co-Chair — July 2024
- Sample Science

- In-Situ Resource Utilization

Lunar and Planetary Science Conference Mar 2023
- Moderator: Planetary Volcanism: Eruptions in Fire and Ice
- Moderator: Lunar Regolith Properties and Processes

SSERVI NASA Exploration Science Forum, Session Co-Chair July 2022
- Apollo Next Generation Sample Analysis

SSERVI NASA Exploration Science Forum, Science Organization Committee Co-Chair Mar 2022 – July 2022

Lunar Surface Science Workshop #17 - Defining a Coordinated Lunar Resource Evaluation Campaign July 2022
- Documentarian

Lunar Surface Science Workshop #13 - Inclusive Lunar Exploration Jan 2022
- Session Co-Chair
- Documentarian

Peer Review Assignments: NASA Research Proposals 2016 – present
- Several NASA Research Opportunities in Space and Earth Science (ROSES) program proposals, once or twice per year. Served as External Reviewer, Executive Secretary, Panelist, and Panel Chair.

Peer Review Assignments: Manuscripts 2015 – present
- International scientific journals, such as *Journal of Geothermal Research, Frontiers in Earth Science, Journal of Volcanology and Geophysical Research, Earth and Planetary Science Letters, Icarus, American Ceramic Society*

Peer-reviewed Publications

Subsurface Temperatures, Thermal Gradient, and Thermal Wave Propagation Depth Constrained Using Thermoluminescence of the Apollo 17 Deep Drill Core 2026
A Sehlke, D Sears, HH Schmitt, and the ANGSA Science Team
https://doi.org/10.1029/2026JE009720 (JGR Planets, Volume 131, Issue 9)

Geomorphological Evidence of Near-Surface Ice at Candidate Landing Sites in Northern Amazonis Planitia, Mars 2025
E Luzzi, JL Heldmann, KE Williams, G Nodjoumi, A Deutsch, ***A Sehlke***
https://doi.org/10.1029/2024JE008724 (JGR Planets, Vol 130(5))

A detailed µ-FTIR study of Hermean glasses: Spectral mainband shape and flank, what do they tell us? 2025
A. Stojic, ***A. Sehlke***, AG Whittington, A. Morlok, H. Hiesinger
https://doi.org/10.1016/j.jnoncrysol.2025.123523 (Journal of Non-Crystalline Solids, Volume 660)

Thermoluminescence and Apollo 17 ANGSA lunar samples: NASA's fifty-year experiment and prospecting for cold traps 2024
DWG Sears, ***A. Sehlke***, HH Schmitt, and the ANGSA Science Team
https://doi.org/10.1029/2024JE008358 (Journal of Geophysical Research: Planets, Volume 129(4))

Apollo Next Generation Sample Analysis (ANGSA): An Apollo Participating Scientist Program to Prepare the Lunar Sample Community for Artemis 2024
CK Shearer, FM McCubbin, S Eckley, SB Simon, A Meshik, F McDonald, HH Schmitt, RA Zeigler, J Gross, J Mitchell, C Krysher, RV Morris, R Parai, BL Jolliff, JJ Gillis-Davis, K Joy, SK Bell, P Lucey, L Sun, Z Sharp, C Dukes, ***A Sehlke***, A Mosie, J Allton, C Amick, JI Simon, TM Erickson, JJ Barnes, MD Dyar, K Burgess, N Petro, D Moriarty, NM Curran, JE Elsila, RA Colina-Ruiz, T Kroll, D Sokaras, HA Ishii, JP Bradley, D Sears, B Cohen, O Pravdivseva, MS Thompson, CR Neal, R Hanna, R Ketcham, K Welten, and the ANGSA Science Team
https://doi.org/10.1007/s11214-024-01094-x (Space Science Review, Volume 220:62)

Synthetic analogs for lava flows on the surface of Mercury: A mid-infrared study 2024
A Morlok, ***A Sehlke***, AN Stojic, AG Whittington, I Weber, MP Reitze, Hiesinger H, Helbert J.
https://doi.org/10.1016/J.ICARUS.2024.116078 (Icarus, Volume 415)

The Apollo 17 Regolith: Induced Thermoluminescence Evidence for Formation by a Single Event ~100 Million Years Ago and Possibly the Presence of Tycho Material 2024
A Sehlke, DWG Sears, and the ANGSA Science Team
https://doi.org/10.1029/2023JE008083 (Journal of Geophysical Research: Planets, Volume 129(4))

Average VNIR reflectance: A rapid, sample-free method to estimate glass content and crystallinity of fresh basaltic lava 2022
E Rader, S Ackiss, ***A Sehlke***, J Bishop, B Orrill, K Odegaard, M Meier, A Doloughan
https://doi.org/10.1016/j.icarus.2022.115084 (Icarus, Volume 383)

Ejecta blocks around the Kings Bowl phreatomagmatic crater in Idaho: An indication of subsurface water amounts with implications for Mars 2022
DWG Sears, ***A Sehlke***, SS Hughes, S Kobs-Nawotniak
10.1016/j.pss.2022.105564 (Planetary and Space Science, Volume 222)

Spontaneous reheating of crystallizing lava 2021
AG Whittington, ***A Sehlke***
10.1130/g49148.1 (Geology, Volume 49 Issue 12)

The impact and recovery of asteroid 2018 LA 2021
P Jenniskens, M Gabadirwe, QZ Yin, A Proyer, O Moses, and with '***A Sehlke***' among 61 other international co-authors'
https://doi.org/10.1111/maps.13653 (Meteoritics & Planetary Science, Volume 56, Issue 4)

Induced thermoluminescence as a method for dating recent volcanism: The Blue Dragon flow, Idaho, USA and the factors affecting induced thermoluminescence 2021
DWG Sears, ***A Sehlke***, SS Hughes
https://doi.org/10.1016/j.pss.2020.105129 (Planetary and Space Science, Volume 195)

Basaltic fissure types on Earth: Suitable analogs to evaluate the origins of volcanic terrains on the Moon and Mars? 2020
SS Hughes, WB Garry, ***A Sehlke***, EH Christiansen, SE Kobs Nawotniak, DWG Sears, RC Elphic, DS Lim, JL Heldmann
10.1016/J.PSS.2020.105091 (Planetary and Space Science, Volume 193)

Thermal properties of glassy and molten planetary candidate lavas 2020
A Sehlke, AM Hofmeister, AG Whittington
https://doi.org/10.1016/J.PSS.2020.105089 (Planetary and Space Science, Volume 193)

Rheology of a KREEP Analog Magma: Experimental Results Applied to Dike Ascent through the Lunar Crust 2020
A Sehlke, AG Whittington
https://doi.org/10.1016/j.pss.2020.104941 (Planetary and Space Science, Volume 187)

Microbial community distribution in variously altered basalts: Insights into astrobiology sample site selection 2020
A Brady, E Gibbons, ***A Sehlke***, C Renner, S Kobs Nawotniak, D Lim, G Slater
10.1016/j.pss.2020.105107 (Planetary and Space Science, Volume 194)

The rheology of crystallizing basaltic lavas from Nyiragongo and Nyamuragira volcanoes, D.R.C. 2020
A Morrison, AG Whittington, B Smets, M Kervyn, ***A Sehlke***
https://doi.org/10.30909/vol.03.01.0128 (Volcanica, Volume 3 Issue 1)

A Low-Diversity Microbiota Inhabits Extreme Terrestrial Basaltic Terrains and Their Fumaroles: Implications for the Exploration of Mars 2019
C Cockell, J Harrison, A Stevens, S Payler, S Hughes, S Kobs Nawotniak, A Brady, R Elphic, C Haberle, ***A Sehlke***, K Beaton, A Abercromby, P Schwendner, J Wadsworth, H Landenmark, R Cane, A Dickinson, N Nicholson, L Perera, D Lim
10.1089/ast.2018.1870 (Astrobiology, Volume 19 Issue 3)

Opportunities and Challenges of Promoting Scientific Dialog throughout Execution of Future Science-Driven Extravehicular Activity 2019
SE Kobs Nawotniak, MJ Miller, AH Stevens, JJ Marquez, SJ Payler, AL Brady, SS Hughes, CW Haberle, ***A Sehlke***, KH Beaton, SP Chappell, RC Elphic, DSS Lim
10.1089/ast.2018.1901 (Astrobiology, Volume 19 Issue 3)

The BASALT Research Program: Designing and Developing Mission Elements in Support of Human Scientific Exploration of Mars 2019
D Lim, A Abercromby, S Kobs Nawotniak, D Lees, M Miller, A Brady, Z Mirmalek, ***A Sehlke***, S Payler, A Stevens, C Haberle, K Beaton, S Chappell, S Hughes, C Cockell, R Elphic, M Downs, JL Heldmann
10.1089/ast.2018.1869 (Astrobiology, Volume 19 Issue 3)

Basaltic Terrains in Idaho and Hawai'i as Planetary Analogs for Mars Geology and Astrobiology 2019
S Hughes, C Haberle, S Kobs Nawotniak, ***A Sehlke***, W Garry, R Elphic, S Payler, A Stevens, C Cockell, A Brady, JL Heldmann, D Lim
10.1089/ast.2018.1847 (Astrobiology, Volume 19 Issue 3)

Strategic Planning Insights for Future Science-Driven Extravehicular Activity on Mars 2019
A Brady, S Kobs Nawotniak, S Hughes, S Payler, A Stevens, C Cockell, R Elphic, ***A Sehlke***, C Haberle, G Slater, D Lim
10.1089/ast.2018.1850 (Astrobiology, Volume 19 Issue 3)

Requirements for Portable Instrument Suites during Human Scientific Exploration of Mars 2019

A Sehlke, Z Mirmalek, D Burtt, CW Haberle, D Santiago-Materese, SE Kobs Nawotniak, SS Hughes, WB Garry, N Bramall, AJ Brown, JL Heldmann, DSS Lim
10.1089/ast.2018.1841 (Astrobiology, Volume 19 Issue 3)

Induced thermoluminescence as a method for dating recent volcanism: Hawaii County, Hawaii, USA 2018
DWG Sears, H Sears, ***A Sehlke***, SS Hughes
10.1016/j.jvolgeores.2017.09.022 (Journal of Volcanology and Geothermal Research, Volume 349)

X-ray computed tomography of extraterrestrial rocks eradicates their natural radiation record and the information it contains 2018
DWG Sears, ***A Sehlke***, JM Friedrich, ML Rivers, DS Ebel
10.1111/maps.13183 (N/A, Volume 53 Issue 12)

Induced thermoluminescence as a method for dating recent volcanism: Eastern Snake River Plain, Idaho, USA 2017
DWG Sears, H Sears, ***A Sehlke***, S Hughes
10.1002/2016JB013596 (Journal of Geophysical Research: Solid Earth, Volume 122 Issue 2)

Transport properties of glassy and molten lavas as a function of temperature and composition 2016
A Hofmeister, ***A Sehlke***, G Avard, A Bollasina, G Robert, AG Whittington
10.1016/j.jvolgeores.2016.08.015 (Journal of Volcanology and Geothermal Research, Volume 327)

The viscosity of planetary tholeiitic melts: A configurational entropy model 2016
A Sehlke, AG Whittington
10.1016/j.gca.2016.07.027 (Geochimica et Cosmochimica Acta, Volume 191)

Field and experimental constraints on the rheology of arc basaltic lavas: the January 2014 Eruption of Pacaya (Guatemala) 2016
A Soldati, ***A Sehlke***, G Chigna, AG Whittington
10.1007/s00445-016-1031-6 (Bulletin of Volcanology, Volume 78 Issue 6)

Rheology of lava flows on Mercury: An analog experimental study 2015
A Sehlke, AG Whittington
10.1002/2015JE004792 (Journal of Geophysical Research E: Planets, Volume 120 Issue 11)

Pahoehoe to a'a' transition of Hawaiian lavas: An experimental study 2014
A Sehlke, AG Whittington, B Robert, AJ Harris, L Gurioli, E Médard
10.1007/s00445-014-0876-9 (Bulletin of Volcanology, Volume 76 Issue 11)

Thermal diffusivity of Fe-rich pyroxene glasses and their melts 2014
A Hofmeister, ***A Sehlke***, AG Whittington
10.1016/j.chemgeo.2014.06.018 (Chemical Geology, Volume)

Textural and rheological evolution of basalt flowing down a lava channel 2014
B Robert, AJ Harris, L Gurioli, E Médard, ***A Sehlke***, AG Whittington
10.1007/s00445-014-0824-8 (Bulletin of Volcanology, Volume 76 Issue 6)

Selected Abstracts and Presentations

Thermochronometry of Lunar Cold Traps via Thermoluminescence: Probing Their Thermal Equilibrium Over Billions of Years 2025

A Sehlke, and DWG Sears
(Goldschmidt Conference 2025, to be held in Prague from July 7-11 2025, Czech Republic - Oral Presentation)

Transit Times of Lunar Meteorites Via Thermoluminescence 2025
J Baden, ***A Sehlke***, and DWG Sears
(56th Lunar and Planetary Science Conference, Abstract Nr. 1672 - Poster Presentation)

Glimmerings in the Cold and Dark: Thermoluminescence of Lunar Regolith at Cryogenic Temperatures for Cold Trap Prospecting 2024
A Sehlke, and DWG Sears
(NASA Exploration Science Forum held at Washington University in St. Louis, MO, USA - Oral Presentation)

Science, Operations, and Technology Development from NASA's RESOURCE Project 2024
JL Heldmann, ***A Sehlke***, MC Deans, and the RESOURCE Team
(NASA Exploration Science Forum held at Washington University in St. Louis, MO, USA - Oral Presentatation)

Update on the Near Infrared Volatiles Spectrometer System (NIRVSS) Instrument on the Volatiles Investigating Polar Exploration Rover (VIPER) Mission: Calibration and Surface Operations 2024
A Sehlke, A Colaprete, K Ennico-Smith, S Gyalay, E Noe Dobrea, TL Roush, JE Benton, R Bielawski, M Chin, J Connally, A Cook, L Ellingson, JB Forgione, DT Hoang, V Jha, A Rademacher, F Renema, EJ Talle, B White, C Youngquist, and the VIPER Science Team
(NASA Exploration Science Forum held at Washington University in St. Louis, MO, USA - Oral Presentation)

The Apollo 17 Regolith: Induced Thermoluminescence Evidence for Formation by a Single Event 100 Million Years Ago and Possibly the Presence of Tycho Material 2024
A Sehlke, DWG Sears, ANGSA Science Team
(55th Lunar and Planetary Science Conference, Abstract Nr. 2536 - Poster Presentation)

The Effect of Composition on the Spectral Appearance of Hermean Analog Glasses 2024
AN Stojic, ***A Sehlke***, A Morlok, AG Whittington, MP Reitze, I Weber, H Hiesinger, J Helbert
(55th Lunar and Planetary Science Conference, Abstract Nr. 1857 - Poster Presentation)

New Model to Calculate Lava Viscosity During Disequilibrium Crystallization for a Wide Range in Cooling and Strain Rates 2023
A Sehlke, AG Whittington
(54th Lunar and Planetary Science Conference, Abstract Nr. 2677 - Poster Presentation)

Lunar Regolith Thermoluminescence Glow Curve Fitting to Extract Its Most Important Kinetic Parameters. 2023
A Sehlke, DWG Sears, ANGSA Science Team
(54th Lunar and Planetary Science Conference, Abstract Nr. 1870 - Oral Presentation)

Geomorphical Evidence of Near-Surface Ice at Candidate Landing Sites in Arcadia Planitia, Mars 2023
E Luzzi, JL Heldmann, K Williams, A Deutsch, ***A Sehlke***, G Nodjoumi
(54th Lunar and Planetary Science Conference, Abstract Nr. 2677 - Oral Presentation)

Thermal Histories of Lunar Cold Traps: Prospecting for Volatiles by Thermoluminescence 2022
A Sehlke, DWG Sears
(Lunar Polar Volatiles Conference, LPI Contrib. Nr. 5024 - Poster Presentation)

A Fifty-Year Experiment, the Natural TL Kinetics of Apollo 17 Regolith, and Prospecting for Water and Other Volatiles on the Moon 2022
A Sehlke, DWG Sears, ANGSA Science Team
(53rd Lunar and Planetary Science Conference, LPI Contrib. Nr. 2030 - Oral Presentation)

THEIA - A Thermal History Exploration Instrument for Artemis 2022
A Sehlke, DWG Sears, JL Heldmann
(Annual Meeting of the Lunar Exploration Analysis Group, Abstract Nr. 5005 - Poster Presentation)

In-Situ Thermoluminescence Measurements on the Moon Using THEIA - Thermal History Exploration Instrument for Artemis 2022
A Sehlke, DWG Sears, JL Heldmann
(NASA Exploration Science Forum held at the University of Colorado, Boulder CO, USA - Poster Presentation)

Crystallization, latent heat release, and thermal history of magmas 2022
AG Whittington, ***A Sehlke***, B Halvernon
(Goldschmidt Conference - Oral Presentation)

Recalescence during crystallization of stardust: Resolution of the amorphous interstellar medium paradox 2022
A Speck, AG Whittington, ***A Sehlke***
(American Astronomical Society Meeting '#240', American Astronomical Society Meeting Abstracts '#147.04' - Poster Presentation)

Five Decades of Thermoluminescence Studies on Lunar Samples: First Results of NASA's Unique 46-Year Experiment and Implications for Resource Prospecting on the Moon 2022
A Sehlke, DWG Sears, ANGSA Science Team
(53rd Lunar and Planetary Science Conference, Abstract Nr. 1267 - Oral Presentation)

High-Temperature Rheology Measurements on Planetary Analog Magmas and Lavas 2022
A Sehlke
(53rd Lunar and Planetary Science Conference, Abstract Nr. 1171 - Poster Presentation)

Lunar Cold Traps: Prospecting by Thermoluminescence 2022
A Sehlke, DWG Sears, ANGSA Science Team
(NASA Exploration Science Forum & European Lunar Symposium, held virtually - Oral Presentation)

Lava Surface Roughness and Morphologies: A New Remote-Sensing Method To Estimate Physical Properties of Lava Flows on Earth, the Moon and Mars 2021
A Sehlke, J Leija, SE Kobs Nawotniak, SS Hughes, DWG Sears, WB Garry, AG Whittington, DSS Lim, JL Heldmann
(Workshop on Terrestrial Analogs for Planetary Exploration, Abstract Nr. 2595 - Oral Presentation)

Natural Thermoluminescence of Lunar Samples: Review and Update 2021
A Sehlke, DWG Sears, ANGSA Science Team

(52nd Lunar and Planetary Science Conference, Abstract Nr. 2548 - Oral Presentation)

A luminescence-based Instrument to Explore the History and Nature of the Lunar Surface 2020
A Sehlke, DWG Sears
(American Geophysical Union 2020 Fall Meeting. Abstract Nr. V013-0006 - Poster Presentation)

Looking Backwards to Look Forward: A Fifty-Year Experiment in the Kinetics of Thermoluminescence of Lunar Samples and the Apollo Next Generation Sample Analysis Program (ANGSA) 2020
A Sehlke, DWG Sears, ANGSA Science Team
(51st Lunar and Planetary Science Conference, Abstract Nr. 1148 - Oral Presentation)

Thermal Properties of Glassy and Molten Planetary Candidate Lavas. 2019
A Sehlke, AM Hofmeister, AG Whittington
(American Geophysical Union, Annual Fall Meeting 2019, Abstract ID V43D-0107 - Poster Presentation)

Requirements for portable instrument suites during human scientific exploration of Mars 2018
A Sehlke, Z Mirmalek, D Burtt, CW Haberle, D Santiago-Materese, SE Kobs Nawotniak, SS Hughes, WB Garry, N Bramall, AJ Brown, JL Heldmann, DSS Lim
(NASA Exploration Science Forum held at NASA Ames Research Center, Moffett Field CA, USA - Oral Presentation)

Synchrotron X-Ray Computed Microtomography and the Radiation History of Meteorites 2017
A Sehlke, DWG Sears, JM Friedrich, ML Rivers, DS Ebel
(80th Annual Meeting of the Meteoritical Society, Abstract Nr. 1987 - Oral Presentation)

The Ultimate Geologic Tricorder? Handheld Science Instruments and Requirements for Future Human Exploration Missions on Other Worlds 2017
A Sehlke, Z Mirmalek, B Cohen, CW Haberle, SE Kobs Nawotniak, SS Hughes, A Brown, JL Heldmann, DSS Lim
(48th Lunar and Planetary Science Conference, Abstract Nr. 2451 - Poster Presentation)

The Viscosity of Tholeiitic Planetary Melts: A Configurational Entropy Model 2016
A Sehlke, AG Whittington
(47th Lunar and Planetary Science Conference, Abstract Nr. 1957 - Poster Presentation)

Rheology of lava flows on Mercury: an experimental study 2015
A Sehlke, AG Whittington
(45th Lunar and Planetary Science Conference, Abstract Nr. 2275 - Poster Presentation)

Concentric cylinder viscometry at subliquidus conditions on Mauna Ulu lavas, Kilauea Volcano, Hawaii 2013
A Sehlke, B Robert, AJ Harris, L Gurioli, AG Whittington
(American Geophysical Union, Annual Fall Meeting 2013, Abstract ID V51D-2697 - Poster Presentation)

Invited Talks, Lectures and Presentations

Requirements for Handheld VNIR and XRF Instruments during Human Exploration Missions. 2022
Lunar Petrology and Landed Instrument Interchange Workshop

(NASA Jet Propulsion Laboratory, Pasadena CA, USA)

Exploration of our Solar System: Earth-based Science Investigations in Preparation for NASAs Moon to Mars Campaign. Graduate Student Seminar. 2021
Graduate Student Seminar (virtual)
(Department of Geological Science, University of Texas at San Antonio, USA)

Rheological and Thermal Evolution of Magmatic Systems: Insights into the Volcanic Past of our Solar System. 2020
Speaker Seminar Series
(Departement of Earth and Planetary Sciences, University of California - Santa Cruz CA, USA)

Rheological and Thermal Evolution of Magmatic Systems: Insights into the Volcanic Past in our Solar System 2019
Keynote at GeoMünster Conference
(Münster, Germany)

Anatomy of the Blue Dragon: Changes in Lava Flow Morphology and Physical Properties Observed in an Open Channel Lava Flow as a Planetary Analogue. 2017
Geological Society of America Annual Meeting
(Seattle WA, USA)

Designing Future Human Spaceflight 2017
Keynote at Sensors Expo 2017
(San Jose CA, USA)

Straight outta morphologies: Understanding the magmatic history of lava terrains on Earth and other rocky worlds in our Solar System 2017
Speaker Seminar Series
(Department of Geological Sciences, San Jose State University, San Jose CA, USA)

The morphological transition from pāhoehoe to aā of basaltic lavas: Combining field studies and experimental work to interpret the volcanic past on Earth and other planets and moons. 2017
Speaker Series Seminar
(United States Geological Survey (USGS), Menlo Park CA, USA)

Technical Reports and other publications (not peer-reviewed)

Inclusive Lunar Exploration 2022
K Bennett and P Prem (Co-Chairs), C Ahrens, N Kumari, L Pigue, ***A Sehlke***, and C Tai Udovicic
(Lunar Surface Science Workshop Virtual Session 13 (LSSW-13) Report)

White Paper: Investigations Regarding Subsurface Temperature Profiles at Polar Regions on the Moon 2020
A Sehlke, and DWG Sears
(Artemis III Science Definition Team Report, White Paper #2107)

Interviews and mentions in Media

The Secrets of Moondust July 2019
written by Marina Koren (The Atlantic Staff Writer)
(The Atlantic)

NASA's lassoing of moon's potential for future use starts with a trove of rocks Mar 2019
written by Peter Fimrite (Science and Environment Reporter)
(San Francisco Chronicle (Front Page, continued on pg. A8))

11 Secrets of Volcanologists Mar 2018
written by Lela Nargi
(Mental Floss)

Research Awards and Funding

The Vertical Gun Range at NASA Ames Oct 2024 – Sept 2028
- NASA ROSES Planetary Science Enabling Facilities Program
- $1,567,144
- PI: Chuck Cornelison, Science PI: Alexander Sehlke

1-Year Funded Extension, Thermoluminescence Studies on Frozen Apollo 17 Samples: Temperature Estimates of Shaded and Illuminated Lunar Surfaces. Apr 2023 – Mar 2024
- NASA ROSES Science Mission Directorate Single-Source
- By invitation only (2022)
- $113,000
- PI: Alexander Sehlke, Co-I: Derek WG Sears

1-Year Funded Extension, Thermoluminescence Studies on Frozen Apollo 17 Samples: Temperature Estimates of Shaded and Illuminated Lunar Surfaces. Apr 2022 – Mar 2023
- NASA ROSES Apollo Next Generation Sample Analysis (ANGSA) Program
- $103,359
- PI: Alexander Sehlke, Co-I: Derek WG Sears

THEIA: Thermal History Exploration Instrument for Artemis Oct 2020 – Sept 2021
- NASA Ames Research Center Innovation Funds
- $41,000
- PI: Jennifer Heldmann, Co-Is: Alexander Sehlke, Derek WG Sears

Resources for Exploration & Science of OUR Cosmic Environment (RESOURCE) Feb 2020 – Jan 2025
- NASA Solar System Exploration Virtual Research Institute (SSERVI)
- $7,452,467
- PI: Jennifer Heldmann, Deputy PIs: Alexander Sehlke, Matthew Deans, Co-Is: 16 across academia, federal agencies and private sector

Thermoluminescence Studies on Frozen Apollo 17 Samples: Temperature Estimates of Shaded and Illuminated Lunar Surfaces. Apr 2019 – Mar 2022
- NASA ROSES Apollo Next Generation Sample Analysis (ANGSA) Program
- $348,050
- PI: Alexander Sehlke, Co-I: Derek WG Sears

Fast and/or Furious? Nature and Emplacement History of Lavas Erupted on Mars May 2019 – Apr 2022
- NASA ROSES Solar System Workings (SSW) Program
- $490,705
- PI: Alexander Sehlke, Co-I: Alan G Whittington

IceCrystal: Portable instrument protocol to delineate ancient ice and water on Mars using microcrystallinity of volcanic products Oct 2018 – Sept 2021

- NASA ROSES Planetary Science and Technology from Analog Research (PSTAR) Program
- $874,012
- PI: Erika Rader, Co-Is: Alexander Sehlke, Janice Bishop

NASA Postdoctoral Fellowship 3rd-Year Extension Feb 2018 – Jan 2019

- NASA ROSES Postodoctoral Program (NPP)
- $86,866
- PI: Alexander Sehlke, Co-Is: Jennifer Heldmann, Darlene SS Lim

NASA Postdoctoral Fellowship Feb 2016 – Jan 2018

- NASA ROSES Postodoctoral Program (NPP)
- $149,248
- PI: Alexander Sehlke, Co-Is: Jennifer Heldmann, Darlene SS Lim

Teaching Experience

University of Missouri - Columbia MO, USA 2011 – 2015

- 2015 Course Instructor – The Moon. Undergraduate
- 2015 Teaching Assistant – Regional Geology Field Course. Undergraduate
- 2014 Teaching Assistant/Lab Experiments – Igneous Petrology. Graduate
- 2013 Teaching Assistant – Mineralogy. Undergraduate
- 2012 Teaching Assistant – Mineralogy. Undergraduate
- 2011 Teaching Assistant – Mineralogy. Undergraduate

Mentoring Experience

NASA Internship mentoring at NASA Ames Research Center

- Summer 2024 (virtual), Arjun Prem, Summit Tahoma Highschool, San Jose CA, USA
- Summer 2024, Jordan Baden, Undergraduate at University of California, Santa Cruz CA, USA
- Summer 2023, Adriana Ariza Pardo, Graduate at UT San Antonio TX, USA
- Spring 2021, Iyare Oseghae. Undergraduate at UT San Antonio TX, USA
- Fall 2020 (virtual), Brianna Orrill, Undergraduate at Arizona State University AZ, USA
- Fall 2020 (virtual), Javier Leija, Undergraduate at Sam Houston University TX, USA
- Summer 2018, Caleb Renner, Undergraduate at Idaho State University ID, USA
- Spring 2017, David Burtt, Undergraduate at Whitman College WA, USA

Technology and Inventions

THEIA - Thermal History Exploration Instrument for Artemis: Instrument protoype to enable thrmoluminescence measurements on the lunar surface via robotic or human exploration missions. Technology Readiness Level (TRL) is 4. Invention is submitted to NASA's *New Technology Reporting (NTR) System*, with e-NTR Number 1684365045.

Public Outreach and Engagement

ExMASS (Exploration of the Moon and Asteroids by Secondary Students) Science Advisor Oct 2023 – July 2024

- Logos Charter School, Medford OR, USA
- virtual

ExMASS (Exploration of the Moon and Asteroids by Secondary Students) Science Advisor Oct 2022 – July 2023

- Logos Charter School, Medford OR, USA
- virtual

Silicon Valley Comic Con, San Jose CA, USA July 2019
- Panel Discussion on *The Artemis Generation: NASA's Journey Forward to the Moon.*

Fremont Peak Observatory, San Juan Bautista CA, USA Aug 2017
- Volcanism on terrestrial planets and moons across our solar system